

GAYA SINGH

singhgaya031@gmail.com — +91-9102088930 — Bhopal, India

GitHub — LinkedIn — Portfolio — Codolio

PROFESSIONAL SUMMARY

Backend Developer specializing in high-performance distributed systems and production-grade REST APIs. Proven expertise in reducing system latency by **42%** and maintaining **99% uptime** through optimized Redis caching and modular architecture. Highly proficient in **Node.js**, **FastAPI**, and **PostgreSQL**, with a focus on building AI-integrated pipelines and scalable data modeling.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL.
- **Backend:** Node.js, Express.js, FastAPI, RESTful APIs, JWT, RBAC.
- **Databases & Caching:** PostgreSQL, MongoDB, Redis, Elasticsearch.
- **Cloud & DevOps:** Render, Vercel, Git, Postman, Cloudinary.
- **Tools:** Zod, LLM, Prompt Engineering, SQLAlchemy.

PROJECTS

Cortex — AI-Powered Code Intelligence Editor — *Next.js, Node.js, MongoDB, Redis* Feb 2026 – July 2026
[Source Code](#) — [Live Demo](#)

- Built an AI-powered Project Intelligence Editor that indexes GitHub repositories and generates Dependency, Call, and Project Flow Graphs to visualize **10,000+ code relationships**.
- Developed AI repository intelligence features including Code Explanation, Repository Chat, Impact Analysis, and Architecture Insights, reducing **reducing manual code exploration by 70%+**.
- Integrated Monaco Editor, intelligent Rename Refactoring, and **Redis caching**, improving repository analysis performance by **up to 5×** for repeated operations.
- **Integrated GitHub repository import and automated indexing**, enabling developers to analyze and explore projects **without local setup**.

RushCart — High-Performance E-commerce Backend — *FastAPI, PostgreSQL, ES, Node.js* Dec 2025 – Jan 2026
[Source Code](#) — [Live Demo](#)

- Cut **auth response time** so that 95% of requests finish in **0.3–0.7 seconds** by optimizing **JWT refresh and Redis session lookup** on FastAPI endpoints.
- Handled **20–60 requests per second** on core read APIs by structuring **async SQLAlchemy queries** and utilizing **response-lean DTOs**.
- Maintained **search availability at ~99%** by implementing **Elasticsearch primary** with a **SQL fallback** and graceful failover mechanism.
- Pushed delivery updates at **12 events/min per partner** by streaming **Socket.IO location events** via a dedicated Node.js microservice.

Karigar — Dual-Role Hyperlocal Services Marketplace — *Node.js, MongoDB, Cloudinary* Oct 2025 – Nov 2025
[Source Code](#) — [Live Demo](#)

- Engineered a **scalable backend** supporting **15+ user flows** across customer, worker, and admin operations using a modular Node.js architecture.
- Designed **7+ REST API modules** and **6 normalized data models** for complex job lifecycle management, dispute resolution, and digital wallet transactions.
- Built a complete **job execution pipeline** covering **5 lifecycle stages** from request creation to closure, including automated review capture and warranty-aware states.
- Developed monetization and proof-of-service features including a **negative wallet with credit limits**, lead-fee refunds, and **3-stage media upload** support.

ACHIEVEMENTS

- **Winner:** 2X National-level Hackathon **Winner**; developed full-stack solution in **<48 hours**.
- **Winner:** National-level Technical Paper Presentation (**Inforia 2K24**).
- **Finalist:** 7+ National-level Hackathons, recognized for **Backend Architecture** excellence.

EDUCATION

IES College of Technology
Bachelor of Technology in Computer Science and Engineering

Expected June 2027
CGPA: 7.31/10.0